Detect Suspicious Activity Using ML

CompTIA SecAI+ — Lesson 1.1.6 | Hands-On Demo

This lesson walks through how Large Language Models can be applied directly to security operations from parsing raw log data to correlating multi-stage attack indicators in real time.

Large Language Models for Security

Security teams today have access to a diverse ecosystem of LLMs each with distinct strengths depending on your environment and threat analysis requirements.

☁️ Cloud-Based Models

Examples: **GPT-4o**, Gemini, Claude

- Trained on vastly larger datasets
- Higher reasoning and correlation capability
- Best for complex, multi-stage analysis
- Requires sending data to external APIs

🖥️ Local Models

Examples: **Mistral**, Phi-3, LLaMA

- Run entirely on-premises or air-gapped
- No data leaves your environment
- Ideal for sensitive or regulated security data
- Smaller parameter counts limit reasoning depth

What Makes a Model More Capable?

Two primary factors determine how sophisticated an LLM's analysis will be in a security context:

Number of Parameters

Parameters are the learned weights inside the model. More parameters generally means the model can hold more nuanced patterns critical for recognizing subtle attack behaviors across long log sequences.

Training Data Quality

A model trained on high-quality, security-relevant data — including CVE databases, threat intelligence, and attack playbooks will produce more accurate and actionable insights than a general-purpose model.

💡 **Key Insight:** For security analysis, model sophistication is not just about size — data quality and domain relevance matter equally.

Analyzing Log Data with AI

Raw security logs contain enormous volumes of data. AI dramatically accelerates triage by identifying patterns that would take analysts hours to spot manually.

Provide Rich Context

Don't just paste a log file — tell the model **what type of data it is** (e.g., Apache access logs, Windows Event Log, firewall output) and **what you're looking for** (e.g., lateral movement, data exfiltration, brute-force attempts). Context dramatically improves output quality.

Basic vs. Advanced Pattern Recognition

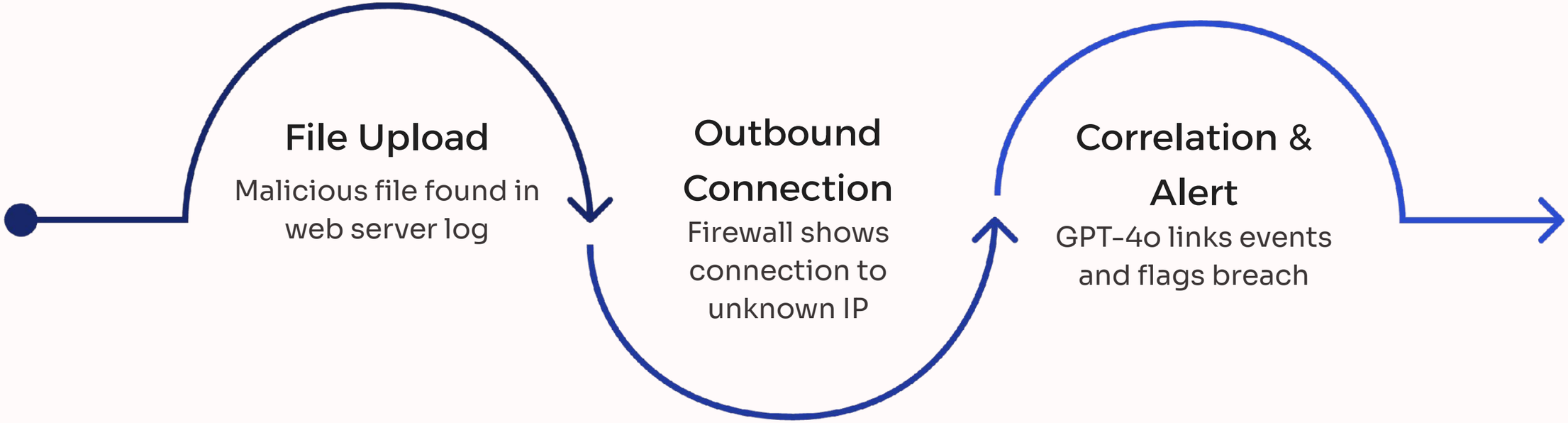
Simpler models may flag individual suspicious entries in isolation — like a single failed login. Advanced models go further, recognizing that **repeated failures followed by a success followed by unusual outbound traffic** is a credential-stuffing attack chain.

Iterative Refinement

Start with broad questions ("What looks suspicious?") and then drill down ("Explain the relationship between these three events"). AI analysis improves with follow-up prompting.

Identifying Breach Indicators

Advanced models like GPT-4o don't just flag events but can also **correlate multiple log entries** across time and systems to reconstruct the attacker's kill chain.



In this example, a file upload event and a subsequent outbound connection, each potentially innocuous alone are combined by the AI to identify a likely **web shell deployment followed by command-and-control callback**. This is the type of multi-stage reasoning that separates advanced models from basic ones, and that saves analysts critical response time during active incidents.

Why Correlation Matters

The Analyst's Challenge

Modern attacks rarely trigger a single alert. Adversaries deliberately spread their activity across time, systems, and log sources to stay below detection thresholds. A SOC analyst manually reviewing thousands of events per day may miss the thread that connects them.
AI acts as a **force multiplier** — ingesting all relevant context simultaneously and surfacing the narrative hidden within the noise.

- Reduces mean time to detect (MTTD)
- Reduces analyst alert fatigue
- Surfaces attack intent, not just raw events

76%

of breaches span multiple systems

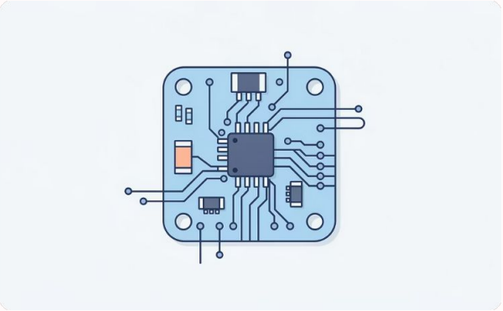
287

avg. days to identify a breach

Without AI-assisted detection

Comparing Model Capabilities

Choosing the right model for a security task is a deliberate decision that balances **analytical depth** against **data sensitivity**. Use this framework to guide your selection:



Basic / Local Models

- Flag individual suspicious events
- Good for pattern matching on known signatures
- Best when data cannot leave the network
- Examples: Mistral 7B, Phi-3 Mini

Advanced Cloud Models

Understand **relationships** between events

- Infer attacker intent and technique
- Best for complex, multi-source investigations
- Examples: GPT-4o, Claude 3.5 Sonnet

🔒 **Decision Rule:** If data contains PII, PHI, or classified content — use a local model. If the analysis involves complex attack reconstruction — use a cloud model on sanitized or non-sensitive data.

Practical Security Use Cases

LLMs are already being applied across the security operations workflow from vulnerability management to incident response. Here are the highest-value applications for practitioners today.

AI in Action: Four Core Use Cases

1

Vulnerability Scan Prioritization

Feed raw scan output (e.g., from Nessus or OpenVAS) into an LLM and ask: *"What are the top 3 critical vulnerabilities I should patch first?"* The model factors in CVSS scores, exploitability, and asset context to produce an actionable ranked list — cutting triage time from hours to minutes.

2

CVE Identification in Logs

Provide application or system logs and ask the model to identify which CVEs may be actively exploited in your environment. The LLM cross-references behavior patterns against known vulnerability signatures, surfacing exploitation attempts you may have missed.

3

Multi-Source Incident Summaries

When an incident involves data from SIEM alerts, endpoint telemetry, and network captures, an LLM can synthesize all sources into a coherent incident narrative — dramatically accelerating post-incident reporting and root cause analysis.

4

Interactive Threat Q&A

Rather than manually reading through large scan files, analysts can ask natural language questions directly: *"Are there any unauthenticated remote code execution vulnerabilities in this report?"* This conversational interface makes deep analysis accessible to analysts at every experience level.

Hands-On Demo: Sample Prompting Strategy

Effective AI-assisted log analysis starts with a well-structured prompt. Here's a battle-tested template you can adapt in your environment:

```
SYSTEM: You are a cybersecurity analyst specializing in log
analysis and threat detection.

USER: I am providing you with [Apache/Windows Event/Firewall]
log data from [date range]. Please:

1. Identify any suspicious or anomalous activity
2. Note relationships between related events
3. Flag any indicators of compromise (IOCs)
4. Summarize your findings as: CRITICAL / HIGH / MEDIUM / LOW
5. Recommend immediate response actions

[PASTE LOG DATA BELOW]
---
```

💡 **Pro Tip:** Always specify the log source type, timeframe, and your analytical goal before pasting data. This context dramatically improves the quality and relevance of AI-generated findings.

Key Takeaways

Choose Your Model Wisely

Match cloud vs. local models to your data sensitivity and analytical complexity requirements.

Context Is Everything

Always give the AI context about the log type, timeframe, and what you're investigating. Vague prompts yield vague results.

Correlation Over Isolation

Advanced models shine when connecting dots across multiple events — use them for complex, multi-stage attack analysis.

AI Accelerates, Analysts Decide

LLMs are force multipliers — they surface insights faster, but human judgment remains essential for final response decisions.

📘 **Next Up — Lesson 1.1.7:** Automating Threat Intelligence Enrichment with AI Pipelines